

Translated

From: e-maxx.ru

k -th order statistic in $O(N)$

Given an array A of size N and a number k . The problem is to find k -th largest number in the array, i.e., k -th order statistic.

The basic idea - to use the idea of quick sort algorithm. Actually, the algorithm is simple, it is more difficult to prove that it runs in an average of $O(N)$, in contrast to the quick sort.

Implementation (not recursive)

```
template <class T>
T order_statistics (std::vector<T> a, unsigned n, unsigned k)
{
    using std::swap;
    for (unsigned l=1, r=n; ; )
    {
        if (r <= l+1)
        {
            // the current part size is either 1 or 2, so it is easy to find
the answer
            if (r == l+1 && a[r] < a[l])
                swap (a[l], a[r]);
            return a[k];
        }

        // ordering a[l], a[l+1], a[r]
        unsigned mid = (l + r) >> 1;
        swap (a[mid], a[l+1]);
        if (a[l] > a[r])
            swap (a[l], a[r]);
        if (a[l+1] > a[r])
            swap (a[l+1], a[r]);
        if (a[l] > a[l+1])
            swap (a[l], a[l+1]);

        // performing division
        // barrier is a[l + 1], i.e. median among a[l], a[l + 1], a[r]
        unsigned
            i = l+1,
            j = r;
        const T
            cur = a[l+1];
        for (;;)
    }
```